

NYC Neighborhood Quality Index

An interactive geospatial analytics platform for exploring neighborhood conditions across New York City using public datasets, GPU-accelerated processing, and dynamic spatial visualization.

The project combines environmental, education, and public safety datasets into a unified **Neighborhood Quality Index (NQI)** that allows users to compare NYC communities through interactive maps, rankings, dashboards, and district-level analysis tools.

Built as a senior design project focused on:

- geospatial data engineering
 - GPU-accelerated analytics
 - interactive visualization
 - neighborhood-level spatial analysis
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Features

- Interactive NYC choropleth map visualization
 - Multiple geography views:
 - Community Districts
 - Boroughs
 - Neighborhood Tabulation Areas (NTAs)
 - ZIP Codes
 - District comparison mode with side-by-side metrics
 - Neighborhood detail pages with rankings and score breakdowns
 - Results dashboard with category analysis and citywide rankings
 - GPU-accelerated crime-data processing using RAPIDS cuDF and CuPy
 - Area-weighted geographic score rollups across multiple NYC boundary systems
 - Dynamic metric-based visualization for:
 - Air Quality
 - Drinking Water
 - Education
 - Public Safety
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Project Highlights

- Built a full-stack geospatial analytics application using Flask, GeoPandas, Leaflet, and Chart.js
- Unified multiple NYC Open Data datasets into a shared neighborhood-level scoring framework
- Processed and normalized heterogeneous datasets with different geographic structures and data formats

- Implemented GPU-accelerated processing pipelines using RAPIDS cuDF and CuPy
 - Reduced large-scale crime pipeline runtime from approximately **2 minutes 13 seconds (CPU)** to **4-5 seconds (GPU)** on server hardware
 - Designed an interactive frontend for neighborhood exploration, ranking analysis, and district comparison
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Screenshots

Landing Page

The landing page introduces the Neighborhood Quality Index platform and provides access to the interactive map and district analysis tools.

Interactive NYC Map

Interactive choropleth visualization showing neighborhood quality scores across New York City. Users can dynamically switch between scoring categories and geography views while exploring district-level conditions.

Compare Mode

Side-by-side district comparison mode allowing users to compare neighborhood performance across air quality, drinking water, education, and public safety metrics.

Results Dashboard

Citywide ranking dashboard displaying category leaders, borough averages, score distributions, and comparative neighborhood analysis.

District Detail Page

Detailed district-level breakdown page displaying category scores, ranking information, and rental platform integrations for practical neighborhood exploration.

What The Neighborhood Quality Index Measures

The Neighborhood Quality Index combines four major categories into a unified scoring framework.

Air Quality

Uses pollutant measurements including:

- PM2.5
- Nitrogen Dioxide (NO₂)
- Ozone (O₃)

Environmental indicators are normalized and aggregated into a composite AirScore.

Drinking Water

Uses public water monitoring indicators including:

- chlorine residual
- turbidity
- fluoride concentration
- bacterial measurements

Sampling-site reconstruction and coordinate enrichment were used to integrate spatially-aware water quality analysis.

Education

Uses district and school-level educational quality indicators transformed from qualitative DOE ratings into standardized numerical scores.

Examples include:

- WD (Well Developed)
 - P (Proficient)
 - D (Developing)
 - U (Underdeveloped)
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Public Safety

Uses NYPD complaint data processed through weighted severity analysis.

Crime incidents were weighted using:

- offense severity
- legal classification
- attempted vs completed status

Higher SafetyScores represent lower overall crime burden.

Neighborhood Quality Index (NQI)

The final Neighborhood Quality Index score is computed as the equal average of the four normalized category scores:

$$\text{NQI} = (\text{Air Quality} + \text{Water Quality} + \text{Education} + \text{Public Safety}) / 4$$

All category scores are normalized into a standardized 0–100 range to allow comparison between datasets with different units and scales.

Processing Pipeline

The system transforms fragmented NYC public datasets into a unified neighborhood-level analysis framework through a multi-stage processing pipeline.

```
Raw NYC Open Data
↓
Data Cleaning & Validation
↓
Feature Transformation
↓
Geospatial Alignment
↓
Z-Score Normalization
↓
Composite Score Generation
↓
Interactive Visualization
```

The pipeline integrates:

- spatial joins
- coordinate reconstruction

- statistical normalization
 - weighted aggregation
 - GPU-accelerated dataframe operations
 - dynamic frontend visualization
-

Geospatial Processing

One of the primary challenges of the project was integrating datasets reported across incompatible geographic systems.

Examples included:

- UHF42 health districts
- police precincts
- water sampling locations
- school-level coordinate data

GeoPandas and GeoJSON boundary processing were used to align all datasets into a shared neighborhood framework supporting:

- Community Districts
- NTAs
- Boroughs
- ZIP Codes

The architecture was designed to support flexible geographic rollups and reusable spatial aggregation.

GPU Acceleration

GPU acceleration was selectively applied to computationally intensive workflows using:

- RAPIDS cuDF
- CuPy
- CUDA-enabled processing

The largest performance gains occurred during processing of the NYPD complaints dataset containing millions of incident-level records.

GPU acceleration enabled:

- parallel aggregation
- weighted incident processing
- normalization

- large-scale dataframe operations

Benchmark

Processing Mode	Runtime
CPU Processing	~2 minutes 13 seconds
GPU Processing	~4–5 seconds

This hybrid CPU/GPU design allowed the project to remain portable while scaling efficiently for large urban datasets.

Tech Stack

Backend & Processing

- Python
- Flask
- pandas
- GeoPandas
- NumPy
- RAPIDS cuDF
- CuPy

Frontend

- HTML
- CSS
- JavaScript
- Leaflet
- Chart.js

Geospatial & Visualization

- GeoJSON
 - Spatial joins
 - Choropleth rendering
 - Interactive dashboards
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Repository Structure

```
frontend/
|
├─ app.py
├─ templates/
└─ static/

scripts/
|
├─ air_quality_process.py
├─ water_quality_process.py
├─ education_quality_process.py
├─ safety_quality_process.py
├─ safety_cd_aggregate.py
└─ build_combined_cd_scores.py

data/
|
└─ geographies/

outputs/
|
└─ combined_nqi_cd_table.csv
```

Running The Project

Clone the Repository

```
git clone <repo-url>
cd nyc-neighborhood-quality-index
```

Install Dependencies

```
pip install -r requirements.txt
```

Run the Application

```
python frontend/app.py
```

Open:

```
http://127.0.0.1:5000
```

Data Sources

The project integrates public datasets from:

- NYC Open Data
 - NYC Department of Environmental Protection
 - NYC Department of Education
 - NYPD Complaint Data
 - NYC Air Quality datasets
 - NYC geographic boundary datasets
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Technical Challenges

Geographic Inconsistency

Datasets used incompatible spatial reporting systems including precincts, UHF42 districts, school locations, and water sampling sites. Spatial alignment required coordinate reconstruction, polygon joins, and area-weighted geographic rollups.

Large-Scale Crime Processing

The NYPD complaints dataset introduced significant computational demands due to its scale and transformation complexity. GPU acceleration was introduced to improve aggregation and normalization performance.

Non-Uniform Data Types

Several datasets contained:

- categorical ratings
- missing coordinates

- inconsistent formatting
- non-numeric measurement values

Custom preprocessing and transformation pipelines were developed to standardize all datasets into comparable scoring frameworks.

Future Improvements

- Improve installation and deployment workflow
 - Add loading states during geography switching
 - Add automated tests for APIs and score pipelines
 - Add historical trend analysis
 - Add filtering and advanced search tools
 - Deploy the application publicly
 - Add real-time or scheduled dataset refreshes
 - Expand neighborhood recommendation functionality
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Contributors

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Project Status

Completed senior design project demonstrating geospatial analytics, public-data integration, GPU-accelerated processing, and interactive neighborhood visualization across New York City.